

Homework 2

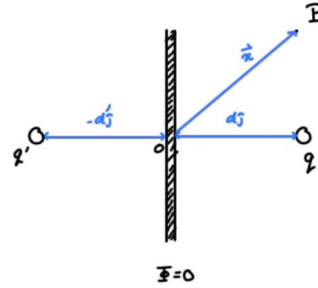
Electromagnetism - PHYS 715

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2.1 A point charge q is brought to a position a distance d away from an infinite plane conductor held at zero potential. Using the method of images, find:

- the surface-charge density induced on the plane, and plot it;
- the force between the plane and the charge by using Coulomb's law for the force between the charge and its image;
- the total force acting on the plane by integrating $\sigma^2/2\epsilon_0$ over the whole plane;
- the work necessary to remove the charge q from its position to infinity;
- the potential energy between the charge q and its image [compare the answer to part d and discuss].
- Find the answer to part d in electron volts for an electron originally one angstrom from the surface.



a) Let the plane be on the xz -plane and q be at $(0, d, 0)$ & q' be at $(0, -d, 0)$.

$$\Phi(\vec{r}) = \frac{q/4\pi\epsilon_0}{|\vec{r}-d\hat{j}|} + \frac{q'/4\pi\epsilon_0}{|\vec{r}+d\hat{j}|}$$

Let $\vec{r}$ be on the xz -plane such that it is on the plane $\Phi=0$.

$$\Phi(\vec{r}) = \frac{q/4\pi\epsilon_0}{r^2+d^2} + \frac{q'/4\pi\epsilon_0}{r^2+d^2} = 0 \quad \vec{r} \cdot \hat{j} = 0 \quad \text{since } \vec{r} \perp \hat{j}$$

$$\boxed{q' = -q, \quad d' = -d}$$

$$\therefore \Phi(\vec{r}) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{|\vec{r}-d\hat{j}|} - \frac{1}{|\vec{r}+d\hat{j}|} \right) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{x^2+(y-d)^2+z^2}} - \frac{1}{\sqrt{x^2+(y+d)^2+z^2}} \right)$$

$$\sigma_{in} = -\epsilon_0 \frac{\partial \Phi}{\partial n} \Big|_{x=0} = -\epsilon_0 \frac{\partial \Phi}{\partial y} \Big|_{y=0} = \frac{-q}{4\pi} \left(\frac{-\frac{1}{2}(2y-d)}{\sqrt{x^2+(y-d)^2+z^2}^3} - \frac{-\frac{1}{2}(2y+d)}{\sqrt{x^2+(y+d)^2+z^2}^3} \right) \Big|_{y=0}$$

$$= \frac{-q}{4\pi (r^2+d^2)^{3/2}} (-2d) = \frac{qd}{2\pi (r^2+d^2)^{3/2}} = \frac{qd}{2\pi (x^2+z^2+d^2)^{3/2}}$$

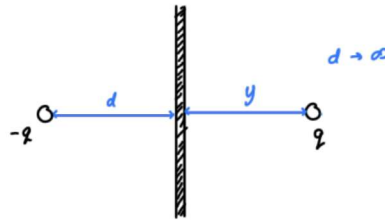
$r^2 = x^2 + z^2$
Since $\vec{r}$ is on the xz plane.

$$b) |\vec{F}| = \frac{|qq'|}{4\pi\epsilon_0 (2d)^2} = \frac{q^2}{16\pi\epsilon_0 d^2}$$

$$\begin{aligned} c) |\vec{F}| &= \frac{1}{2\epsilon_0} \iint_{xz \text{ plane}} \sigma^2 dA \\ &= \frac{1}{2\epsilon_0} \int_0^{2\pi} \int_0^\infty \left[\frac{qd}{2\pi (r^2+d^2)^{3/2}} \right]^2 r dr d\theta \\ &= \frac{q^2 d^2}{8\pi\epsilon_0} \int_0^\infty \frac{2r}{(r^2+d^2)^3} dr \\ &= \frac{q^2 d^2}{8\pi\epsilon_0} \int_{d^2}^\infty \frac{1}{u^3} du \quad \begin{matrix} u = r^2 + d^2 \\ du = 2r dr \end{matrix} \\ &= \frac{q^2 d^2}{8\pi\epsilon_0} \left[\frac{u^{-2}}{-2} \right]_{d^2}^\infty = \frac{q^2}{16\pi\epsilon_0 d^2} \end{aligned}$$

$$\begin{aligned}
 d) \quad W &= \int_d^\infty F \, dy \\
 &= \int_d^\infty \frac{q^2}{4\pi\epsilon_0 (y+d)^2} \, dy \\
 &= \frac{q^2}{4\pi\epsilon_0} \int_d^\infty \frac{1}{(y+d)^2} \, dy \\
 &= \frac{q^2}{4\pi\epsilon_0} \int_{2d}^\infty u^{-2} \, du \\
 &= \frac{q^2}{4\pi\epsilon_0} \left. \frac{u^{-1}}{-1} \right|_{2d}^\infty \\
 &= \frac{q^2}{8\pi\epsilon_0 d}
 \end{aligned}$$

$$\begin{aligned}
 u &= y+d \\
 du &= dy
 \end{aligned}$$



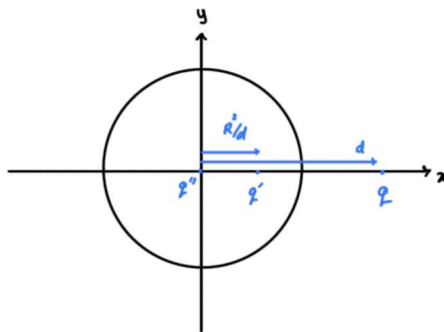
$$e) \quad \text{Potential Energy} = Fd = \frac{q^2}{16\pi\epsilon_0 d} \cdot 2d = \frac{q^2}{8\pi\epsilon_0 d} \quad \text{same as d.}$$

$$\begin{aligned}
 f) \quad W &= \frac{q^2}{8\pi\epsilon_0 d} = \frac{(1 \text{ eV})^2}{8\pi (8.85 \times 10^{-12} \text{ C}^2 \text{ V}^{-1} \text{ m}^{-1}) (10^{-10} \text{ m})} \cdot \frac{1.6 \times 10^{-19} \text{ C}}{1 \text{ eV}} \\
 &\approx 7.2 \text{ eV}
 \end{aligned}$$

2.3 A point charge is placed a distance $d > R$ from the center of an equally charged, isolated, conducting sphere of radius R .

- (a) Inside of what distance from the surface of the sphere is the point charge attracted rather than repelled by the charged sphere?
 (b) What is the limiting value of the force of attraction when the point charge is located a distance $a (= d - R)$ from the surface of the sphere, if $a \ll R$?
 (c) What are the results for parts a and b if the charge on the sphere is twice (half) as large as the point charge, but still the same sign?

[Answers: (a) $d/R - 1 = 0.6180$, (b) $F = -q^2/(16\pi\epsilon_0 a^2)$, i.e., image force, (c) for $Q = 2q$, $d/R - 1 = 0.4276$; for $Q = q/2$, $d/R - 1 = 0.8823$. The answer for part b is the same.]



a) We can solve this problem by assuming a further charge q'' in the centre of the conductor such that $q' + q'' = q$ to account for the conductor charge.

Therefore, we have :

$$q' = -\frac{R}{d} q \quad \text{from (2.2 Jackson)}$$

$$q'' = q - q' = (1 + R/d) q$$

$$|\vec{F}| = \frac{qq'}{4\pi\epsilon_0 (d - R/d)^2} + \frac{qq''}{4\pi\epsilon_0 d^2}$$

$$= \frac{q^2}{4\pi\epsilon_0} \left[\frac{-R/d}{(d - R/d)^2} + \frac{(1 + R/d)}{d^2} \right]$$

For q to be attracted, $|\vec{F}| > 0$.

$$\frac{1 + R/d}{d^2} > \frac{R/d}{(d - R/d)^2}$$

$$(1 + R/d)(d - R/d)^2 - Rd > 0$$

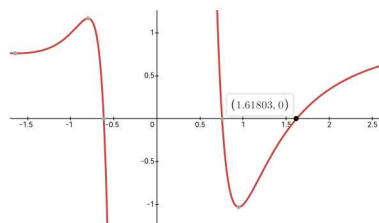
$$(1 + R/d)(1 - R^2/d^2)^2 - R/d > 0 \quad [\text{divide by } 1/d^2]$$

By using desmos, we can find the intervals such that $R/d < 1$ & $f(x) > 0$ such that

$$f(x) = (1 + \frac{1}{x})(1 - \frac{1}{x^2})^2 - \frac{1}{x}$$

There is only one root that satisfies the conditions.

$$x_{\text{root}} = d/R = 1.61803 \dots$$



$$b) |\vec{F}| = \frac{q^2}{4\pi\epsilon_0} \left[\frac{-\frac{R}{d}}{(d - \frac{R^2}{d})^2} + \frac{(1 + \frac{R}{d})}{d^2} \right]$$

$$= \frac{q^2}{4\pi\epsilon_0} \left[\frac{-dR}{(d^2 - R^2)^2} + \frac{1}{d^2} + \frac{R}{d^3} \right]$$

We now replace d with $a+R = R(\frac{a}{R} + 1)$

$$|\vec{F}| = \frac{q^2}{4\pi\epsilon_0} \left[\frac{-R(\frac{a}{R}+1)R}{R^4(\frac{a}{R}+1)^2 - 1)^2} + \frac{1}{R^2(\frac{a}{R}+1)^2} + \frac{1}{R^2(\frac{a}{R}+1)^3} \right]$$

$$= \frac{q^2}{4\pi\epsilon_0} \left[\frac{-(\frac{a}{R}+1)}{R^2(\frac{a}{R})^2(\frac{a}{R}+2)^2} + \frac{1}{R^2(\frac{a}{R}+1)^2} + \frac{1}{R^2(\frac{a}{R}+1)^3} \right]$$

$$= \frac{q^2}{4\pi\epsilon_0} \left[\frac{-(\frac{a}{R}+1)}{a^2(\frac{a^2}{R^2} + 2\frac{a}{R} + 4)} + \frac{1}{R^2(\frac{a}{R}+1)^2} + \frac{1}{R^2(\frac{a}{R}+1)^3} \right]$$

As we are trying to see the limiting behavior for $a \ll R$ i.e. $\lim \frac{a}{R} \rightarrow 0$.

$$\approx \frac{q^2}{4\pi\epsilon_0} \left[\frac{-1}{4a^2} + \frac{2}{R^2} \right]$$

We find that at the limit as $a \rightarrow 0$, the force is dominated by the attractive force caused by the induced charge that acts like an image charge at a distance $2a$.

- c) We are going to investigate how doubling or halving the charge of the conductor affects:
- 1) The limit where the force becomes attractive
 - 2) The limiting behavior of the force $a \rightarrow 0$.

[1] We notice that changing the charge of the conductor only affects the first term of our $f(x)$.

$$q' = -\frac{R}{d} q$$

$$q'' = 2q - q' = (2 + \frac{R}{d}) q$$

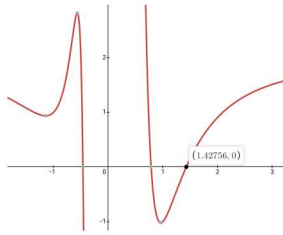
$$q' = -\frac{R}{d} q$$

$$q'' = \frac{1}{2} q - q' = (\frac{1}{2} + \frac{R}{d}) q$$

If we double q :

$$f(x) = (2 + \frac{1}{x}) (1 - \frac{1}{x^2})^2 - \frac{1}{x}$$

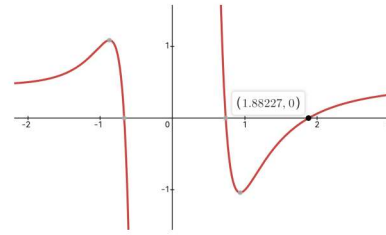
$$x_{\text{root}} = d/R = 1.42756 \dots$$



If we half q :

$$f(x) = (\frac{1}{2} + \frac{1}{x}) (1 - \frac{1}{x^2})^2 - \frac{1}{x}$$

$$x_{\text{root}} = d/R = 1.88227 \dots$$



[2] If we look at the limiting behavior calculations, we find there is no change $F \sim \frac{q^2}{16\pi\epsilon_0 a^2}$.

This makes sense when we think of induced charge becoming more dominant which depends on the charge of the point charge q and not on the charge of the conductor

2.4 (a) Show that the work done to remove the charge q from a distance $r > a$ to infinity against the force, Eq. (2.6), of a grounded conducting sphere is

$$W = \frac{q^2 a}{8\pi\epsilon_0(r^2 - a^2)}$$

Relate this result to the electrostatic potential, Eq. (2.3), and the energy discussion of Section 1.11.

(b) Repeat the calculation of the work done to remove the charge q against the force, Eq. (2.9), of an isolated charged conducting sphere. Show that the work done is

$$W = \frac{1}{4\pi\epsilon_0} \left[\frac{q^2 a}{2(r^2 - a^2)} - \frac{q^2 a}{2r^2} - \frac{qQ}{r} \right]$$

Relate the work to the electrostatic potential, Eq. (2.8), and the energy discussion of Section 1.11.

a)
$$W = - \int_r^\infty |F| dy$$

$$= - \int_r^\infty \frac{1}{4\pi\epsilon_0} \frac{q^2}{a^2} \left(\frac{a}{y} \right)^3 \left(1 - \frac{a^2}{y^2} \right)^{-2} dy$$

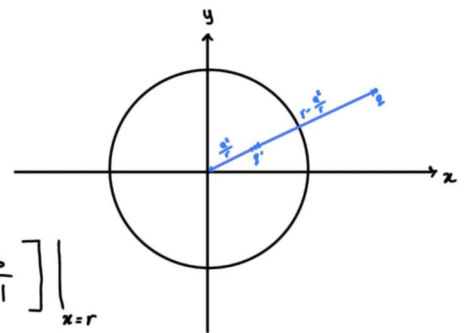
$$= \int_{1-\frac{a^2}{r^2}}^1 \frac{1}{8\pi\epsilon_0} \frac{q^2}{a} u^{-2} du$$

$$= \frac{q^2}{8\pi\epsilon_0 a} \frac{u^{-1}}{-1} \Big|_{1-\frac{a^2}{r^2}}^1$$

$$= \frac{q^2}{8\pi\epsilon_0 a} \left(\frac{r^2}{r^2 - a^2} - 1 \right) = \frac{q^2 a}{8\pi\epsilon_0 (r^2 - a^2)} \quad (1)$$

$$u = 1 - \frac{a^2}{y^2}$$

$$du = -\frac{2a^2}{y^3} dy$$



The electrostatic potential $= -q \Phi(r) = -q \left[\frac{q/4\pi\epsilon_0}{|\vec{x} - \vec{y}|} + \frac{q'/4\pi\epsilon_0}{|\vec{x} - \vec{y}'|} \right] \Big|_{\vec{x}=\vec{r}}$

$$= -\frac{q^2}{4\pi\epsilon_0} \left[0 - \frac{a}{r(r - a^2/r)} \right]$$

$$= \frac{q^2 a}{4\pi\epsilon_0 (r^2 - a^2)} \quad (2)$$

We find that the answers differ by a factor of 2. The classical interpretation of the work done W is taken to be the energy required to move a point to infinity, but we note that the induced charge on the conductor is variable depending on the location of the point charge. We therefore adopt (1) rather than (2) for our W .

b) Calculation of W is quite similar as there is an additional term in the force from Q to account for.

$$q_{in} + q_{reflector} = Q_{total} \Rightarrow q_{reflector} = Q - q_{in} \text{ or } Q - q'$$

$$F_{q_{reflector}} = \frac{q(Q - q')}{4\pi\epsilon_0 y^2}$$

$$F = \frac{1}{4\pi\epsilon_0} \frac{q}{y^2} \left[Q - \frac{qa^3(2y^2 - a^2)}{y(y^2 - a^2)^2} \right] \frac{y}{y}$$

We therefore have ...
$$W = \int_r^\infty F_{q_{reflector}} + F_a^{\text{part a}} dy$$

We solve for the first part.

$$\int_r^\infty F_{qQ_{\text{effective}}} dy = - \int_r^\infty \frac{q(Q-q')}{4\pi\epsilon_0 y^2} dy$$

F is -ve since q & Q have the same charge

$$\begin{aligned} &= \frac{-1}{4\pi\epsilon_0} \int_r^\infty \frac{qQ}{y^2} + \frac{q^2 a}{y^3} dy \\ &= \frac{-1}{4\pi\epsilon_0} \left[-\frac{qQ}{y} - \frac{q^2 a}{2y^2} \right]_r^\infty \\ &= \frac{-1}{4\pi\epsilon_0} \left[\frac{qQ}{r} + \frac{q^2 a}{2r^2} \right] \quad (3) \end{aligned}$$

$$\text{Therefore, } W = (1) + (3) = \frac{1}{4\pi\epsilon_0} \left[\frac{q^2 a}{2(r^2 - a^2)} - \frac{qQ}{r} - \frac{q^2 a}{2r^2} \right]$$

$$\begin{aligned} \text{The electrostatic potential} = -q\bar{\Phi}(r) &= -q \left[\frac{q/4\pi\epsilon_0}{|\vec{x}-\vec{y}|} + \frac{q'/4\pi\epsilon_0}{|\vec{x}-\vec{y}|} + \frac{(Q-q')/4\pi\epsilon_0}{|\vec{x}|} \right] \Big|_{\vec{x}=r} \\ &= \frac{-q}{4\pi\epsilon_0} \left[0 - \frac{aq}{r(r-a^2/r)} + \frac{Q + \frac{q}{r}q}{r} \right] \\ &= \frac{q^2 a}{4\pi\epsilon_0 (r^2 - a^2)} - \frac{q^2 a}{4\pi\epsilon_0 r^2} - \frac{qQ}{4\pi\epsilon_0 r} \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{q^2 a}{(r^2 - a^2)} - \frac{qQ}{r} - \frac{q^2 a}{r^2} \right] \end{aligned}$$

2.6 A two-dimensional potential problem is defined by two straight parallel line charges separated by a distance R with equal and opposite linear charge densities λ and $-\lambda$.

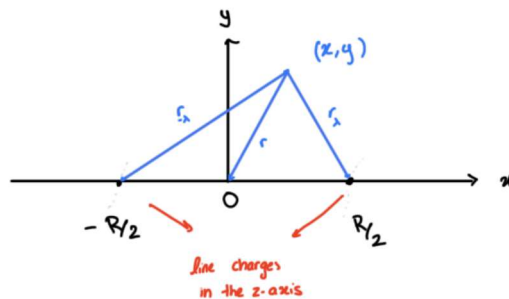
(a) Show by direct construction that the surface of constant potential V is a circular cylinder (circle in the transverse dimensions) and find the coordinates of the axis of the cylinder and its radius in terms of R , λ , and V .

(b) Use the results of part a to show that the capacitance per unit length C of two right-circular cylindrical conductors, with radii a and b , separated by a distance $d > a + b$, is

$$C = \frac{2\pi\epsilon_0}{\cosh^{-1}\left(\frac{d^2 - a^2 - b^2}{2ab}\right)}$$

(c) Verify that the result for C agrees with the answer in [Problem 1.5](#) in the appropriate limit and determine the next nonvanishing order correction in powers of a/d and b/d .

(d) Repeat the calculation of the capacitance per unit length for two cylinders inside each other ($d < |b - a|$). Check the result for concentric cylinders ($d = 0$).



a) Finding E for a line charge :

$$\oint \vec{E} \cdot d\vec{\alpha} = \frac{Q_{in}}{\epsilon_0}$$

$$E(2\pi r l) = \frac{Q_{in}}{\epsilon_0}$$

$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0} \hat{r}$$

Find $\Phi(x, y)$ for line charge λ :

$$\begin{aligned} \Phi(\vec{r}_\lambda) - \Phi(\vec{r}_0) &= -\int_{r_0}^{r_\lambda} E dr \\ &= -\frac{\lambda}{2\pi\epsilon_0} (\ln r_\lambda - \ln r_0) \end{aligned}$$

$$\Phi(x, y) = \Phi(\vec{r}_\lambda) + \Phi(\vec{r}_{-\lambda}) = \frac{\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_{-\lambda}}{r_\lambda}\right)$$

Find $\Phi(x, y)$ for line charge $-\lambda$:

$$\begin{aligned} \Phi(\vec{r}_{-\lambda}) - \Phi(\vec{r}_0) &= -\int_{r_0}^{r_{-\lambda}} E dr \\ &= \frac{\lambda}{2\pi\epsilon_0} (\ln r_\lambda - \ln r_0) \end{aligned}$$

Suppose the Φ is constant i.e. $\Phi(x, y) = V$.

$$\ln\left(\frac{r_{-\lambda}}{r_\lambda}\right) = \frac{2\pi\epsilon_0 V}{\lambda}$$

$$\frac{r_{-\lambda}}{r_\lambda} = \exp\left\{\frac{2\pi\epsilon_0 V}{\lambda}\right\} \equiv K \text{ constant}$$

$$r_{-\lambda}^2 = K^2 r_\lambda^2$$

$$(x + R/2)^2 + y^2 = K^2 (x - R/2)^2 + K^2 y^2$$

$$x^2 + Rx + \frac{R^2}{4} + y^2 = K^2 x^2 - K^2 Rx + K^2 \frac{R^2}{4} + K^2 y^2$$

$$(1 - K^2)x^2 + (1 + K^2)Rx + (1 - K^2)\frac{R^2}{4} + (1 - K^2)y^2 = 0$$

$$x^2 + \underbrace{\frac{(1 + K^2)}{1 - K^2} R}_{b} x + \frac{R^2}{4} + y^2 = 0$$

$$(x - x_0)^2 + y^2 = \rho^2$$

$$r_\lambda = \sqrt{(x - R/2)^2 + y^2}$$

$$r_{-\lambda} = \sqrt{(x + R/2)^2 + y^2}$$

$$x^2 + \frac{bx}{2} + \frac{b^2}{4}$$

$$\left(x + \frac{b}{2}\right)^2$$

$$\text{where } x_0 = -\frac{b}{2} = \frac{R}{2} \frac{k^2+1}{k^2-1} \quad \text{and } \rho^2 = \left(\frac{b}{2}\right)^2 - \frac{R^2}{4} = \frac{R^2}{4} \frac{(1+k^2)^2}{(1-k^2)^2} - \frac{R^2}{4}$$

$$= \frac{R^2}{4} \frac{[(1+k^2) - (1-k^2)][(1+k^2) + (1-k^2)]}{(1-k^2)^2}$$

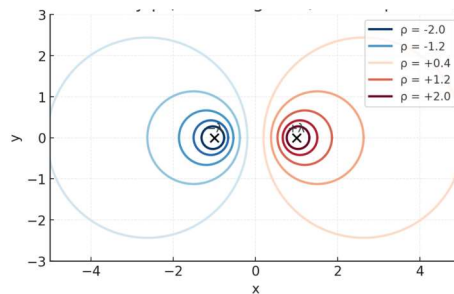
$$= \frac{R^2}{4} \frac{4k^2}{(1-k^2)^2}$$

$$\Rightarrow \rho = R \frac{k}{|1-k^2|}$$

$$\therefore x_0 = \frac{R}{2} \frac{k^2+1}{k^2-1} = \frac{R}{2} \left[\frac{k+k^{-1}}{k-k^{-1}} \right] = \frac{R}{2} \left[\coth \left(\frac{2\pi\epsilon_0 V}{\lambda} \right) \right] \quad (1) \quad k = e^{\frac{2\pi\epsilon_0 V}{\lambda}}$$

$$\rho = \frac{R}{2} \frac{2}{|k^{-1}-k|} = \frac{R}{2} \frac{1}{|\sinh(\frac{2\pi\epsilon_0 V}{\lambda})|} \quad (2)$$

Therefore, surfaces of equipotential are circles with centres $(0, x_0)$ & radii ρ



b) Goal: $C' = \frac{2\pi\epsilon_0}{\cosh^{-1}\left(\frac{d^2-a^2-b^2}{2ab}\right)} = \frac{\lambda}{\Delta V} = \frac{2\pi\epsilon_0}{\ln(r_1/r_2)}$

New Goal: Show that $\ln\left(\frac{r_1}{r_2}\right) = \cosh^{-1}\left(\frac{d^2-a^2-b^2}{2ab}\right)$

proof | Surfaces of conductor are equipotential, so we can use our results from part a by considering the cylindrical conductors as line charges such that:

Cylinder_a has radius a and center x_a } With distance $d = x_a + x_b$
 Cylinder_b has radius b and center x_b } between them

Φ on cylinder_a surface $-V_a$ and Φ on cylinder_b surface V_b
 with line charge $-\lambda$ with line charge λ

Let $J_a = -\frac{2\pi\epsilon_0 V_a}{\lambda}$ and $J_b = \frac{2\pi\epsilon_0 V_b}{\lambda}$.

$$\begin{aligned}\Delta V &= V_b - (-V_a) \\ &= \frac{\lambda}{2\pi\epsilon_0} \left(\frac{2\pi\epsilon_0 V_b}{\lambda} - \frac{2\pi\epsilon_0 V_a}{\lambda} \right) \\ &= \frac{\lambda}{2\pi\epsilon_0} (J_b - J_a)\end{aligned}$$

We have from (1) and (2) that :

	Radii	Centers	Separation
Cylinder a	$a = -\frac{R}{2} \frac{1}{\sinh J_a}$	$x_a = \frac{R}{2} \coth J_a$	$d = x_b - x_a = \frac{R}{2} (\coth J_b - \coth J_a)$
Cylinder b	$b = \frac{R}{2} \frac{1}{\sinh J_b}$	$x_b = \frac{R}{2} \coth J_b$	

We work with further identities.

$$\begin{aligned}\sinh J_a &= -\frac{R}{2a} & \cosh J_a &= \sqrt{1 + \sinh^2 J_a} = \frac{\sqrt{R^2 + 4a^2}}{2a} \\ \sinh J_b &= \frac{R}{2b} & \cosh J_b &= \sqrt{1 + \sinh^2 J_b} = \frac{\sqrt{R^2 + 4b^2}}{2b}\end{aligned}$$

$$\begin{aligned}d &= \frac{R}{2} \left(\frac{\sqrt{R^2 + 4b^2}}{2b} \times \frac{2b}{R} - \frac{\sqrt{R^2 + 4a^2}}{2a} \times -\frac{2a}{R} \right) \\ &= \frac{1}{2} (\sqrt{R^2 + 4a^2} + \sqrt{R^2 + 4b^2}) \\ d^2 &= \frac{1}{4} (R^2 + 4a^2 + R^2 + 4b^2 + 2\sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}) \\ &= \frac{R^2}{2} + a^2 + b^2 + \frac{\sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}}{2}\end{aligned}$$

We finally investigate $\cosh(J_b - J_a)$

$$\begin{aligned}\cosh(J_b - J_a) &= \cosh J_a \cosh J_b - \sinh J_a \sinh J_b \\ &= \frac{\sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}}{4ab} + \frac{R^2}{4ab} \\ &= \frac{R^2 + \sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}}{2} = \frac{d^2 - a^2 - b^2}{2ab}\end{aligned}$$

$$\therefore C' = \frac{\lambda}{\Delta V} = \frac{\lambda}{\frac{\lambda}{2\pi\epsilon_0} (J_b - J_a)} = \frac{2\pi\epsilon_0}{\cosh^{-1} \left(\frac{d^2 - a^2 - b^2}{2ab} \right)}$$

c) We have that

$$C = \frac{2\pi\epsilon_0}{\cosh^{-1}\left(\frac{d^2 - a^2 - b^2}{2ab}\right)}$$

By using the inverse coshine & expanding

$$\begin{aligned}\cosh^{-1}x &= \ln(x + \sqrt{x^2 - 1}) \\ &= \ln(x + x\sqrt{1 - 1/x^2}) \\ &= \ln\left(x + x\left(1 - \frac{1}{2}\left(\frac{1}{x^2}\right) + O\left(\frac{1}{x^4}\right)\right)\right) \\ &\approx \ln(2x) + O\left(\frac{1}{x^2}\right)\end{aligned}$$

$$\text{Let } x = \frac{d^2 - (a^2 + b^2)}{2ab} = \frac{1}{2} \frac{d^2}{ab} \left(1 - \frac{a^2 + b^2}{d^2}\right)$$

$$\begin{aligned}\cosh^{-1}x &\approx \ln(2x) \\ &\approx \ln\left(\frac{d^2}{ab}\left(1 - \frac{a^2 + b^2}{d^2}\right)\right) \\ &\approx \ln\left(\frac{d^2}{ab}\right) + \ln\left(1 - \frac{a^2 + b^2}{d^2}\right) \\ &\approx \underbrace{\ln(d^2/ab)}_L - \underbrace{\frac{a^2 + b^2}{d^2}}_\epsilon\end{aligned}$$

$$\int \frac{1}{1-x} = 1 + x + x^2 + \dots \quad dx$$

$$-\ln(1-x) = x + \frac{x^2}{2} + \frac{x^3}{3} + \dots$$

$$\ln(1-x) \approx -x$$

We finally finish off with C'

$$C' = \frac{2\pi\epsilon_0}{L - \epsilon} = \frac{2\pi\epsilon_0}{L(1 - \epsilon/L)} \approx \frac{2\pi\epsilon_0}{L} \left(1 + \frac{\epsilon}{L}\right) \quad \frac{1}{1-x} = 1 + x + \dots$$

$$\approx \frac{2\pi\epsilon_0}{\ln(d^2/ab)} \left[1 + \frac{\frac{a^2 + b^2}{d^2}}{\ln(d^2/ab)}\right]$$

$$\text{Letting } a=b \Rightarrow C' \approx \frac{2\pi\epsilon_0}{\ln(d^2/a^2)} = \frac{\pi\epsilon_0}{\ln(d/a)} \quad \text{which matches 1.5}$$

d) WLOG (Without loss of Generality) suppose $a > b$.

Then, both $x_a, x_b > 0$ & $J_a, J_b > 0$.

	Radii	Centers	Separation
Cylinder a	$a = \frac{R}{2} \frac{1}{\sinh J_a}$	$x_a = \frac{R}{2} \coth J_a$	$d = x_b - x_a = \frac{R}{2} (\coth J_b - \coth J_a)$
Cylinder b	$b = \frac{R}{2} \frac{1}{\sinh J_b}$	$x_b = \frac{R}{2} \coth J_b$	

We work with further identities.

$$\begin{aligned} \sinh J_a &= \frac{R}{2a} & \cosh J_a &= \sqrt{1 + \sinh^2 J_a} = \frac{\sqrt{R^2 + 4a^2}}{2a} \\ \sinh J_b &= \frac{R}{2b} & \cosh J_b &= \sqrt{1 + \sinh^2 J_b} = \frac{\sqrt{R^2 + 4b^2}}{2b} \end{aligned} \quad \left| \quad \begin{aligned} d &= \frac{R}{2} \left(\frac{\sqrt{R^2 + 4b^2}}{2b} \times \frac{2b}{R} - \frac{\sqrt{R^2 + 4a^2}}{2a} \times \frac{2a}{R} \right) \\ &= \frac{1}{2} (\sqrt{R^2 + 4a^2} - \sqrt{R^2 + 4b^2}) \\ d^2 &= \frac{1}{4} (R^2 + 4a^2 + R^2 + 4b^2 - 2\sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}) \\ &= \frac{R^2}{2} + a^2 + b^2 - \frac{\sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}}{2} \end{aligned} \right.$$

We finally investigate $\cosh(J_b - J_a)$

$$\begin{aligned} \cosh(J_b - J_a) &= \cosh J_a \cosh J_b - \sinh J_a \sinh J_b \\ &= \frac{\sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}}{4ab} - \frac{R^2}{4ab} \\ &= \frac{-R^2 + \sqrt{R^2 + 4a^2}\sqrt{R^2 + 4b^2}}{2} = \frac{a^2 + b^2 - d^2}{2ab} \end{aligned}$$

$$\therefore C' = \frac{\lambda}{\Delta V} = \frac{\lambda}{\frac{\lambda}{2\pi\epsilon_0}(J_b - J_a)} = \frac{2\pi\epsilon_0}{\cosh^{-1}\left(\frac{a^2 + b^2 - d^2}{2ab}\right)}$$

If cylinder are concentric, then $d=0$.

$$\begin{aligned} \therefore \cosh^{-1}\left(\frac{a^2 + b^2}{2ab}\right) &= \ln\left(\frac{a^2 + b^2}{2ab} + \sqrt{\left(\frac{a^2 + b^2}{2ab} + 1\right)\left(\frac{a^2 + b^2}{2ab} - 1\right)}\right) \\ &= \ln\left(\frac{a^2 + b^2 + (a+b)(a-b)}{2ab}\right) = \ln\left(\frac{2a^2}{2ab}\right) = \ln(a/b) \end{aligned}$$

$$C' = \frac{2\pi\epsilon_0}{\cosh^{-1}\left(\frac{a^2 + b^2}{2ab}\right)} = \frac{2\pi\epsilon_0}{\ln(a/b)} \quad \text{where } a > b. \quad \square$$

2.7 An insulated, spherical, conducting shell of radius a is in a uniform electric field E_0 . If the sphere is cut into two hemispheres by a plane perpendicular to the field, find the force required to prevent the hemispheres from separating

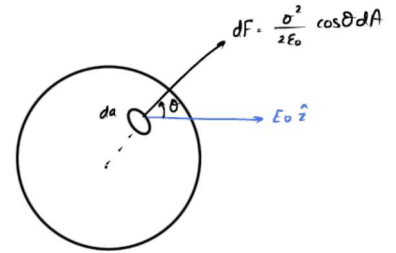
(a) if the shell is uncharged;

(b) if the total charge on the shell is Q .

a) Consider $\Phi(\vec{x})$ &

$$\begin{aligned}\Phi(\vec{x}) &= \Phi_{\text{shell charge}} + \Phi_E \\ &= \frac{Q}{4\pi\epsilon_0 r} - E_0 \left(r - \frac{a^3}{r^2} \right) \cos\theta\end{aligned}$$

$$\begin{aligned}\sigma_n &= -\epsilon_0 \left. \frac{\partial \Phi}{\partial n} \right|_{r=a} = -\epsilon_0 \left. \frac{\partial \Phi}{\partial x} \right|_{x=a} = -\epsilon_0 \left[-\frac{Q}{4\pi\epsilon_0 r^2} - E_0 \left(1 - \frac{2a^3}{r^3} \right) \cos\theta \right] \Big|_{r=a} \\ &= \frac{Q}{4\pi a^2} + 3\epsilon_0 E_0 \cos\theta\end{aligned}$$



Let $dA = a^2 \sin\theta d\theta d\phi$

$$F = \frac{1}{2\epsilon_0} \iint_{\text{hemisphere}} \sigma^2 \cos\theta dA$$

$$\begin{aligned}&= \frac{1}{2\epsilon_0} \int_0^{2\pi} \int_0^{\pi/2} \left[\frac{Q^2}{16\pi^2 a^4} + \frac{3Q\epsilon_0 E_0}{2\pi a^2} \cos\theta + 9\epsilon_0^2 E_0^2 \cos^2\theta \right] \cos\theta \sin\theta d\theta d\phi \\ &= \frac{\pi a^2}{\epsilon_0} \int_0^{\pi/2} \left[\frac{Q^2}{16\pi^2 a^4} + \frac{3Q\epsilon_0 E_0}{2\pi a^2} \cos\theta + 9\epsilon_0^2 E_0^2 \cos^2\theta \right] \cos\theta \sin\theta d\theta\end{aligned}$$

$$\int_0^{\pi/2} \cos\theta \sin\theta d\theta$$

$$= \frac{1}{2} \cos^2\theta \Big|_0^{\pi/2}$$

$$= \frac{1}{2}$$

$$\int_0^{\pi/2} \cos^2\theta \sin\theta d\theta$$

$$= \frac{1}{3} \cos^3\theta \Big|_0^{\pi/2}$$

$$= \frac{1}{3}$$

$$\int_0^{\pi/2} \cos^3\theta \sin\theta d\theta$$

$$= \frac{1}{4} \cos^4\theta \Big|_0^{\pi/2}$$

$$= \frac{1}{4}$$

$$\begin{aligned}F &= \frac{\pi a^2}{\epsilon_0} \left[\frac{Q^2}{32\pi^2 a^4} + \frac{Q\epsilon_0 E_0}{2\pi a^2} + \frac{9}{4} \epsilon_0^2 E_0^2 \right] \\ &= \frac{Q^2}{32\pi\epsilon_0 a^2} + \frac{QE_0}{2} + \frac{9}{4} \pi a^2 \epsilon_0 E_0^2\end{aligned}$$

a) $Q = 0 \Rightarrow F = \frac{9\pi}{4} \epsilon_0 a^2 E_0^2$

b) $Q = Q_{\text{surface}} \Rightarrow F = \frac{Q^2}{32\pi\epsilon_0 a^2} + \frac{QE_0}{2} + \frac{9}{4} \pi a^2 \epsilon_0 E_0^2$

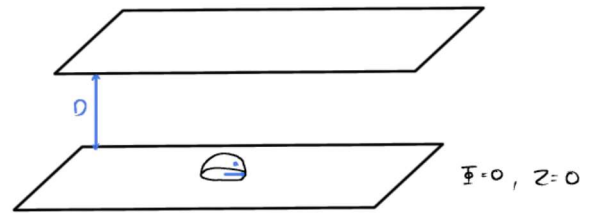
2.8 A large parallel plate capacitor is made up of two plane conducting sheets with separation D , one of which has a small hemispherical boss of radius a on its inner surface ($D \gg a$). The conductor with the boss is kept at zero potential, and the other conductor is at a potential such that far from the boss the electric field between the plates is E_0 .

(a) Calculate the surface-charge densities at an arbitrary point on the plane and on the boss, and sketch their behavior as a function of distance (or angle).

(b) Show that the total charge on the boss has the magnitude $3\pi\epsilon_0 E_0 a^2$.

(c) If, instead of the other conducting sheet at a different potential, a point charge q is placed directly above the hemispherical boss at a distance d from its center, show that the charge induced on the boss is

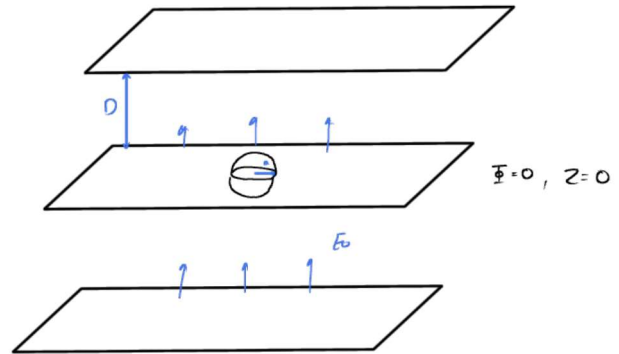
$$q' = -q \left[1 - \frac{d^2 - a^2}{d\sqrt{d^2 + a^2}} \right]$$



We attempt a technique to use what we learned about a grounded sphere in an electric field.

We reflect the whole set up across $z=0$.

We end up getting a sphere at potential $\Phi=0$ in an electric field $E_0 \hat{z}$



a) $\Phi(\vec{r}) = -E_0 \left(r - \frac{a^3}{r^2} \cos\theta \right)$

$$\sigma_{\text{boss}} = 3\epsilon_0 E_0 \cos\theta \quad \text{for } 0 < \theta \leq \pi/2$$

from 2.7.

To find σ_{plane} , it is more convenient to work with cylindrical coordinates such that

$$r = \sqrt{s^2 + z^2} \quad r \cos\theta = z$$

$$\therefore \Phi(s, \theta, z) = -E_0 \left(z - \frac{za^3}{(s^2 + z^2)^{3/2}} \right)$$

$$\begin{aligned} \sigma_{\text{plane}} &= -\epsilon_0 \left. \frac{\partial \Phi}{\partial z} \right|_{z=0^+} = -\epsilon_0 E_0 \left[1 - \frac{\sqrt{s^2 + z^2}^3 \cdot 0^3 - 2a^3(-\frac{1}{2})\sqrt{s^2 + z^2}^5}{(s^2 + z^2)^3} \right] \bigg|_{z=0^+} \\ &= -\epsilon_0 E_0 \left(1 - \frac{a^3}{s^3} \right) \end{aligned}$$

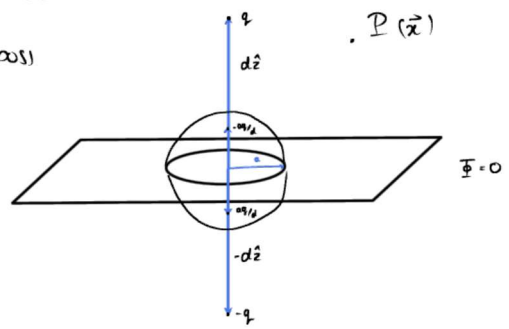
b) Total Charge on boss = $\int_{\text{hemisphere}} dq = \int_0^{2\pi} \int_0^{\pi/2} 3\epsilon_0 E_0 \cos\theta \cdot a^2 \sin\theta \, d\theta \, d\phi$

$$= 3\epsilon_0 E_0 a^2 \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos\theta \sin\theta \, d\theta$$

$$= 3\pi \epsilon_0 E_0 a^2$$

- c) We use the same setup as part b with plates replaced with point charges q at a distance $D+a$ from the centre of the boss and $-q$ at a distance $D+a$ in the opposite direction

$$\therefore \Phi(\vec{x}) = \frac{q/4\pi\epsilon_0}{|\vec{x}-d\hat{z}|} - \frac{q/4\pi\epsilon_0}{|\vec{x}+d\hat{z}|} - \underbrace{\frac{q/d}{|\vec{x}-\frac{d^2}{d}\hat{z}|}}_{\text{image charge from } q} + \underbrace{\frac{q/d}{|\vec{x}+\frac{d^2}{d}\hat{z}|}}_{\text{image charge from } -q}$$



In order to find Q_{boss} , we proceed in the following steps:-

- 1) Find $\sigma_{\text{plane}} = -\epsilon_0 \frac{\partial \Phi}{\partial z} \Big|_{z=0}$
- 2) Find $Q_{\text{plane}} = \int \sigma_{\text{plane}} dA$
- 3) $Q_{\text{boss}} = Q_{\text{image charge}} - Q_{\text{plane}}$

- 1) We work with cylindrical coordinates to simplify calculations.

$$\Phi(s, \theta, z) = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\sqrt{s^2+(z-d)^2}} - \frac{1}{\sqrt{s^2+(z+d)^2}} - \frac{q/d}{\sqrt{s^2+(z-\frac{d^2}{d})^2}} + \frac{q/d}{\sqrt{s^2+(z+\frac{d^2}{d})^2}} \right]$$

$$\begin{aligned} \sigma_{\text{plane}} &= -\epsilon_0 \frac{\partial \Phi}{\partial z} \Big|_{z=0^+} = -\frac{q}{4\pi} \left[\frac{d-z}{\sqrt{s^2+(z-d)^2}^3} + \frac{d+z}{\sqrt{s^2+(z+d)^2}^3} + \frac{q/d(z-\frac{d^2}{d})}{\sqrt{s^2+(z-\frac{d^2}{d})^2}^3} - \frac{q/d(z+\frac{d^2}{d})}{\sqrt{s^2+(z+\frac{d^2}{d})^2}^3} \right]_{z=0^+} \\ &= -\frac{q}{4\pi} \left[\frac{2d}{\sqrt{s^2+d^2}^3} - \frac{2a^3/d^2}{\sqrt{s^2+\frac{a^4}{d^2}}^3} \right] \end{aligned}$$

$$\begin{aligned} 2) \quad Q_{\text{plane}} &= \int_{\text{plane}} \sigma_{\text{plane}} dA \\ &= \int_0^{2\pi} \int_a^\infty \sigma_{\text{plane}} s ds d\theta \\ &= -\frac{q}{2} \int_a^\infty \frac{2ds}{\sqrt{s^2+d^2}^3} - \frac{2a^3/d^2 s}{\sqrt{s^2+\frac{a^4}{d^2}}^3} ds \\ &= -\frac{q}{2} (-2) \int_a^\infty \frac{-\frac{1}{2}(2s) d}{\sqrt{s^2+d^2}^3} - \frac{(-\frac{1}{2})(2s)(\frac{a^3}{d^2})}{\sqrt{s^2+\frac{a^4}{d^2}}^3} ds \\ &= q \left[\frac{d}{\sqrt{s^2+d^2}} - \frac{a^3/d^2}{\sqrt{s^2+\frac{a^4}{d^2}}} \right] \Big|_a^\infty \end{aligned}$$

$$= q \left[\frac{a^3/d^2}{\sqrt{a^2 + a^4/b^2}} - \frac{d}{\sqrt{a^2 + d^2}} \right]$$

$$= q \left[\frac{a^2/d - d}{\sqrt{a^2 + d^2}} \right]$$

$$= q \left[\frac{a^2 - d^2}{d\sqrt{a^2 + d^2}} \right]$$

$$\begin{aligned} 3) \quad Q_{\text{boss}} &= Q_{\text{image charge}} - Q_{\text{plane}} \\ &= -q + \frac{aq}{d} - \frac{aq}{d} - q \left[\frac{a^2 - d^2}{d\sqrt{a^2 + d^2}} \right] \\ &= -q \left[1 - \frac{d^2 - a^2}{d\sqrt{a^2 + d^2}} \right] \end{aligned}$$